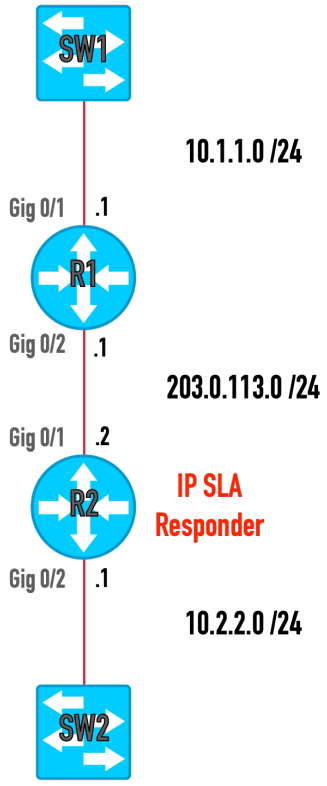
IP SLA - Response Time Reporter (RTR)

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
```

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```
no ip domain-lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
no shut
int gig 0/2
ip address 203.0.113.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 203.0.113.2 255.255.255.0
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

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Lab Tasks

1. To start our configuration, let's go over to R2 and configure it as an IP SLA Responder. Afterwards, let's verify that it is enabled.
2. On R1, create an IP SLA operation with an entry number of "1" that targets UDP jitter and points to the IP address "203.0.113.2" on R2, our IP SLA responder.
3. Continuing with our configuration, configure our voice traffic to use the first available UDP port number, set the codec to G.729A, and specify a codec size of 20.
4. Still on R1, let's configure a ToS byte of 184.
5. On R1, let's send out our probes every 5 seconds.
6. On R1, schedule our probes to be sent now, and initiate the probe sending forever.
7. Let's wait a few minutes to allow the probes to be sent every 5 seconds from R1 to R2. R2 should have responded with some statistical information. We can review the statistics collected by R1 using the "show ip sla statistics" command.

Solution

Step 1: To start our configuration, let's go over to R2 and configure it as an IP SLA Responder. Afterwards, let's verify that it is enabled.

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip sla ?
  <1-2147483647>          Entry Number
  auto                    IP SLAs Auto Configuration
  enable                  Enable Event Notifications
  endpoint-list           Endpoint list configuration
  ethernet-monitor        IP SLAs Auto Ethernet Configuration
  group                   Group Configuration or Group Scheduling
  key-chain               Use MD5 Authentication for IP SLAs Control Messages
  logging                 Enable Syslog
  low-memory              Configure Low Water Memory Mark
  reaction-configuration  IP SLAs Reaction-Configuration
  reaction-trigger        IP SLAs Trigger Assignment
  reset                   IP SLAs Reset
  responder               Enable IP SLAs Responder
  restart                 Restart An Active Entry
  schedule                Entry Scheduling
  server                  IPPM server configuration

R2(config)#ip sla responder
R2(config)#end
R2#show ip sla responder
        General IP SLA Responder on Control port 1967
```

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```
        General IP SLA Responder on Control V2 port 1167
General IP SLA Responder is: Enabled
Number of control message received: 0 Number of errors: 0
Recent sources:
Recent error sources:

Number of control v2 message received: 0 Number of errors: 0
Recent sources:
Recent error sources:

        Permanent Port IP SLA Responder
Permanent Port IP SLA Responder is: Disabled
```

(##Notice hos IP SLA Responder is now enabled on R2.)

Step 2: On R1, create an IP SLA operation with an entry number of "1" that targets UDP jitter and points to the IP address "203.0.113.2" on R2, our IP SLA responder.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip sla ?
  <1-2147483647>          Entry Number
  auto                    IP SLAs Auto Configuration
  enable                  Enable Event Notifications
  endpoint-list           Endpoint list configuration
  ethernet-monitor        IP SLAs Auto Ethernet Configuration
  group                   Group Configuration or Group Scheduling
  key-chain               Use MD5 Authentication for IP SLAs Control Messages
  logging                 Enable Syslog
  low-memory              Configure Low Water Memory Mark
  reaction-configuration  IP SLAs Reaction-Configuration
  reaction-trigger        IP SLAs Trigger Assignment
  reset                   IP SLAs Reset
  responder               Enable IP SLAs Responder
  restart                 Restart An Active Entry
  schedule                Entry Scheduling
  server                  IPPM server configuration

R1(config)#ip sla 1
R1(config-ip-sla)#?
IP SLAs entry configuration commands:
  dhcp         DHCP Operation
  dns          DNS Query Operation
  ethernet     Ethernet Operations
  exit         Exit Operation Configuration
  ftp          FTP Operation
  http         HTTP Operation
  icmp-echo    ICMP Echo Operation
```

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```
  icmp-jitter  ICMP Jitter Operation
  mpls         MPLS Operation
  path-echo    Path Discovered ICMP Echo Operation
  path-jitter  Path Discovered ICMP Jitter Operation
  tcp-connect  TCP Connect Operation
  udp-echo     UDP Echo Operation
  udp-jitter   UDP Jitter Operation
  voip         Voice Over IP Operation

R1(config-ip-sla)#udp-jitter ?
  Hostname or A.B.C.D  Destination IP address or hostname
```

(##This is used for our RTP traffic.)
(##Here is where we point to the IP address of R2.)

Step 3: Continuing with our configuration, configure our voice traffic to use the first available UDP port number, set the codec to G.729A, and specify a codec size of 20.

```
R1(config-ip-sla)#udp-jitter 203.0.113.2 ?
  <1-65535>  Port Number. Recommended ports greater than 1023. For codec
             options recommended port range is an Even Port range between
             16384-32767 or 49152-65535

R1(config-ip-sla)#udp-jitter 203.0.113.2 16384 ?
  codec        codec type to be configured
  control      Enable or disable control packets
  interval     Inter Packet Interval
  num-packets  Number of Packets to be transmitted
  source-ip    Source address
  source-port  Source Port
  <cr>         <cr>

R1(config-ip-sla)#udp-jitter 203.0.113.2 16384 codec ?
  g711alaw  G.711 A Law 64000 bps
  g711ulaw  G.711 U Law 64000 bps
  g729a     G.729 8000 bps

R1(config-ip-sla)#udp-jitter 203.0.113.2 16384 codec g729a ?
  advantage-factor  Advantage Factor
  codec-interval    Inter Packet Interval
  codec-numpackets  Number of Packets to be transmitted
  codec-size        Number of bytes in payload
  control           Enable or disable control packets
  source-ip         Source address
  source-port       Source Port
  <cr>              <cr>

R1(config-ip-sla)#udp-jitter 203.0.113.2 16384 codec g729a codec-size 20
```

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(##Because of concerns with jitter, we're likely on a lower bandwidth link, making it more likely that we would use G.729a due to its lower complexity compression algorithm.)
(##G.729 packets typically feature a payload size of 20 bytes and a L3/L4 header size totaling 40 bytes by default.)

Step 4: Still on R1, let's configure a ToS byte of 184.

```
R1(config-ip-sla-jitter)#?
IP SLAs Udp Jitter Configuration Commands:
  clock-tolerance            Set acceptable clock synchronization error
  control                    Set control protocol parameters
  default                    Set a command to its defaults
  dest-ipaddr                Destination ip address
  dest-port                  Destination port
  exit                       Exit operation configuration
  frequency                  Frequency of an operation
  history                    History and Distribution Data
  no                         Negate a command or set its defaults
  operation-packet-priority  Set operation packet properties
  optimize                   Optimize timestamp of measurement
  owner                      Owner of Entry
  percentile                 Set percentile statistics levels
  precision                  Set precision of measurement
  tag                        User defined tag
  threshold                  Operation threshold in milliseconds
  timeout                    Timeout of an operation
  tos                        Type Of Service
  verify-data                Verify data
  vrf                        Configure IP SLAs for a VPN Routing/Forwarding
                             instance

R1(config-ip-sla-jitter)#tos ?
  <0-255>  Type of Service Value

R1(config-ip-sla-jitter)#tos 184
```

(##We are giving this the same quality of service marking that a voice packet would have.)

Step 5: On R1, let's send out our probes every 5 seconds.

```
R1(config-ip-sla-jitter)#frequency ?
  <1-604800>  Frequency in seconds

R1(config-ip-sla-jitter)#frequency 5
R1(config-ip-sla-jitter)#exit
```

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Step 6: On R1, schedule our probes to be sent now, and initiate the probe sending forever.

```
R1(config)#ip sla schedule ?
  <1-2147483647>  Entry number

R1(config)#ip sla schedule 1 ?
  ageout      How long to keep this Entry when inactive
  life        Length of time to execute in seconds
  recurring   Probe to be scheduled automatically every day
  start-time  When to start this entry
  <cr>        <cr>

R1(config)#ip sla schedule 1 start-time ?
  after     Start after a certain amount of time from now
  hh:mm     Start time (hh:mm)
  hh:mm:ss  Start time (hh:mm:ss)
  now       Start now
  pending   Start pending
  random    Start after random time (range)

R1(config)#ip sla schedule 1 start-time now ?
  ageout     How long to keep this Entry when inactive
  life       Length of time to execute in seconds
  recurring  Probe to be scheduled automatically every day
  <cr>       <cr>

R1(config)#ip sla schedule 1 start-time now life ?
  <0-2147483647>  Life seconds (default 3600)
  forever         continue running forever

R1(config)#ip sla schedule 1 start-time now life forever
R1(config)#exit
```

(##We need to reference our previously-configured IP SLA Operation and use the same entry number of 1.)
(##This indicates that we want to start sending probes now.)
(##This indicates that we want to send probes forever, or at least until we unconfigure it.)

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Step 7: Let's wait a few minutes to allow the probes to be sent every 5 seconds from R1 to R2. R2 should have responded with some statistical information. We can review the statistics collected by R1 using the "show ip sla statistics" command.

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 1
Type of operation: udp-jitter
        Latest RTT: 4 milliseconds
Latest operation start time: 10:33:55 UTC Wed Jul 17 2024
Latest operation return code: OK
RTT Values:
        Number Of RTT: 1000             RTT Min/Avg/Max: 1/4/11 milliseconds
Latency one-way time:
        Number of Latency one-way Samples: 0
        Source to Destination Latency one way Min/Avg/Max: 0/0/0 milliseconds
        Destination to Source Latency one way Min/Avg/Max: 0/0/0 milliseconds
Jitter Time:
        Number of SD Jitter Samples: 999
        Number of DS Jitter Samples: 999
        Source to Destination Jitter Min/Avg/Max: 0/2/8 milliseconds
        Destination to Source Jitter Min/Avg/Max: 0/2/13 milliseconds
Over Threshold:
        Number Of RTT Over Threshold: 0 (0%)
Packet Loss Values:
        Loss Source to Destination: 0
        Source to Destination Loss Periods Number: 0
        Source to Destination Loss Period Length Min/Max: 0/0
        Source to Destination Inter Loss Period Length Min/Max: 0/0
        Loss Destination to Source: 0
        Destination to Source Loss Periods Number: 0
        Destination to Source Loss Period Length Min/Max: 0/0
        Destination to Source Inter Loss Period Length Min/Max: 0/0
        Out Of Sequence: 0      Tail Drop: 0
        Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
        Calculated Planning Impairment Factor (ICPIF): 11
        MOS score: 4.06
Number of successes: 18
Number of failures: 0
Operation time to live: Forever
```

(##Here, based on the statistics we can see our entry number of "1", and that we configured our type of operation for "udp-jitter".)

(##This indicates our most recent round-trip time, which is the duration it took to travel from R1 to R2 and back again.)

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(##This displays the number of packets sent and provides the minimum, average, and maximum round-trip times.)
(##This indicates that our jitter averages 2 milliseconds, which is excellent. The industry standard states that we should not exceed 150 milliseconds.)
(##MOS (Mean Opinion Score) gives us a score as to the quality of our voice.)

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